



BLM4021 Gömülü Sistemler

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Yıldız Teknik Üniversitesi – Bilgisayar Mühendisliği

Sunum 8 – Analog Sensörlerden Veri Toplama ve Dönüştürücüler

- Analog-Dijital Dönüştürme
- Dijital-Analog Dönüştürme
- Önemli Parametreler
- Uygulama Örneği

Hafta	Teorik	Laboratuvar
1	Giriş ve Uygulamalar, Mikroişlemci, Mikrodenetleyici ve Gömülü sistem kavramlarının açıklanması	Grupların oluşturulması & Kitlerin Testi
2	Bir Tasarım Örneği, Mikroişlemci, Mikrodenetleyici, DSP, FPGA, ASIC kavramları	Kitlerin gruplara dağıtımı + Raspberry Pi Kurulumu
3	16, 32 ve 64 bitlik mikrodenetleyiciler, pipeline	Raspberry Pi ile Temel Konfigürasyon
4	PIC ve MSP430 özellikleri	Uygulama 1 — Raspberry Pi ile Buzzer Uygulaması
5	ARM ve RISC-V tabanlı mimariler ve özellikleri	Uygulama 2 — Raspberry Pi ile İvme ve Gyro Uygulaması
6	ARM Komut setleri ve Assembly Kodları 1	Uygulama 3 — Raspberry Pi ile Motor Kontrol Uygulaması
7	ARM Komut setleri ve Assembly Kodları 2, Raspberry Pi vers. ve GPIO'ları	Uygulama 4 — Raspberry Pi ile Görüntü İşleme Uygulaması
8	Vize Sınavı	
9	Çoklu ortam algılayıcıları ve arayüzleri (SPI, I2C...)	Uygulama 5 — Raspberry Pi ile Network Uygulaması
10	Sensörlerden Veri Toplama, Algılayıcı, ADC ve DAC	Proje Soru-Cevap Saati
11	Zamanlayıcı, PWM ve Motor Sürme	Proje kontrolü-1
12	Gerçek Zaman Sistemlerinde temel kavramlar	Proje Kontrolü-2
13	Gerçek zaman İşletim Sistemleri	Mazeret sebepli son proje kontrollerinin yapılması
14	Nesnelerin İnterneti	
15	Final Sınavı	

Gerekli Kaynaklar:

- Derek Molloy, Exploring Raspberry Pi: Interfacing to the Real World with Embedded Linux, Wiley, 2016.
- M. Wolf, Computers as Components: Principles of Embedded Computing System Design, Elsevier, 2008.

Yardımcı Kaynaklar:

- D. Zhu, T. Sifleet, T. Nunnally, Y. Huang, Analog to Digital Converters, Lecture Notes.
- Fast and Effective Embedded System Design: Applying the ARM mbed.
- Farid Farahmand, Chapter- 3, Embedded Systems with ARM Cortex-M, 2018.
- Tolga Ayav, Embedded Control Systems, Data Acquisition and Digital Signal Processing.
- Simon Monk, Raspberry Pi Cookbook, O'Reilly.

Data Converters: Basic Concepts



\$1.2-8 bit



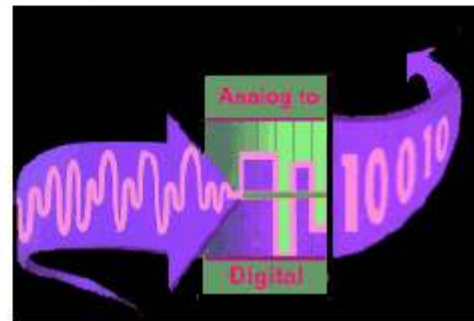
\$48-18 bit

- ❑ Analog signals are continuous, with infinite values in a given range.
- ❑ Digital signals have discrete values such as on/off or 0/1.
- ❑ Limitations of analog signals
 - Analog signals pick up noise as they are being amplified.
 - Analog signals are difficult to store.
 - Analog systems are more expensive in relation to digital systems.

Analog and Digital Signals

- Advantages of digital systems (signals)
 - Noise can be reduced by converting analog signals in 0s and 1s.
 - Binary signals of 0s/1s can be easily stored in memory.
 - Technology for fabricating digital systems has become so advanced that they can be produced at low cost.
- The major limitation of a digital system is how accurately it represents the analog signals after conversion.

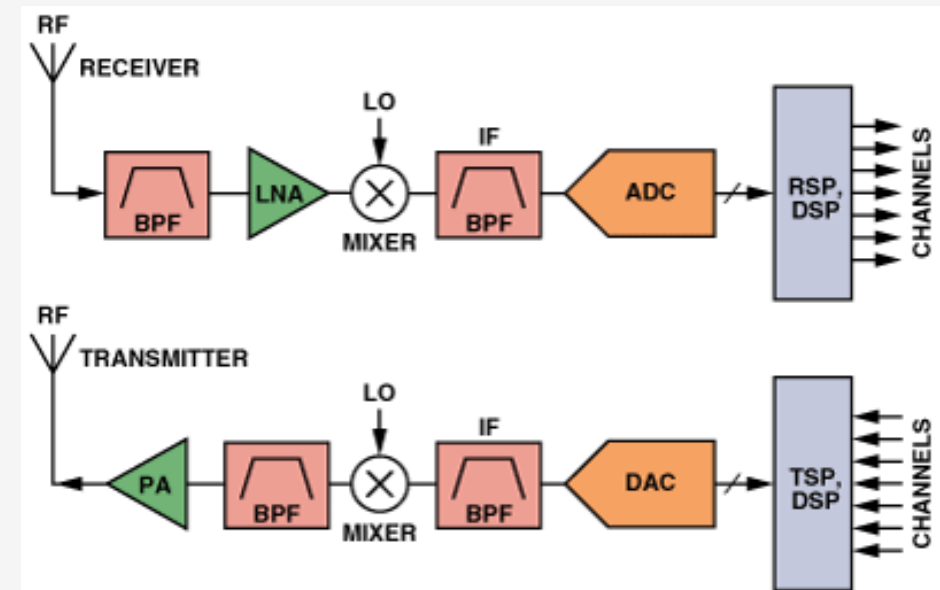
Accuracy is the
key concept!



Application of ADC

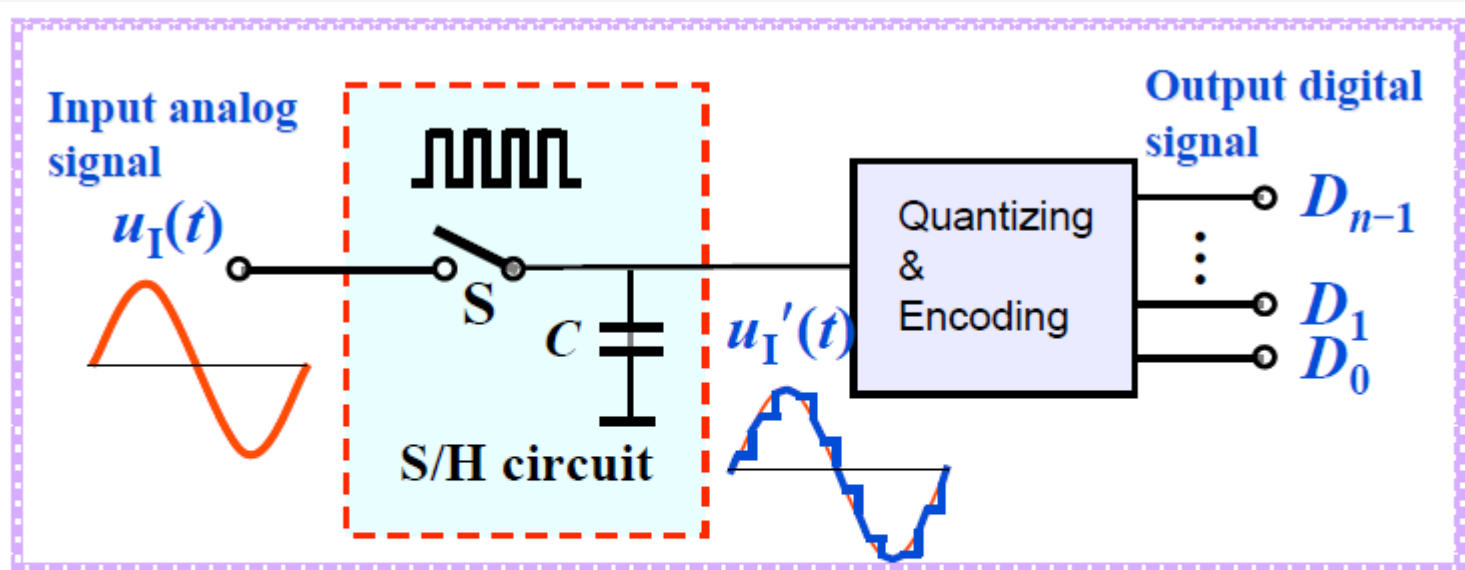
- ADC are used virtually everywhere where an analog signal has to be processed, stored, or transported in digital form.
- Some examples of ADC usage are digital volt meters, cell phone, thermocouples, and digital oscilloscope.
- Microcontrollers commonly use 8, 10, 12, or 16 bit ADCs, our micro controller uses an 8 or 10 bit ADC.

Radio Receiver and Transmitter:



- LNA: Low-noise amplifier
- LO: Local Oscillator

ADC Process



2 steps

- Sampling and Holding (S/H)
- Quantizing and Encoding (Q/E)

İdeal Örnekleme

Örnek: $x(t) = \cos(2\pi 100t)$

Genel form: $x(t) = \cos(2\pi ft)$ $\rightarrow$ Frekans: $f = \frac{1}{T}$

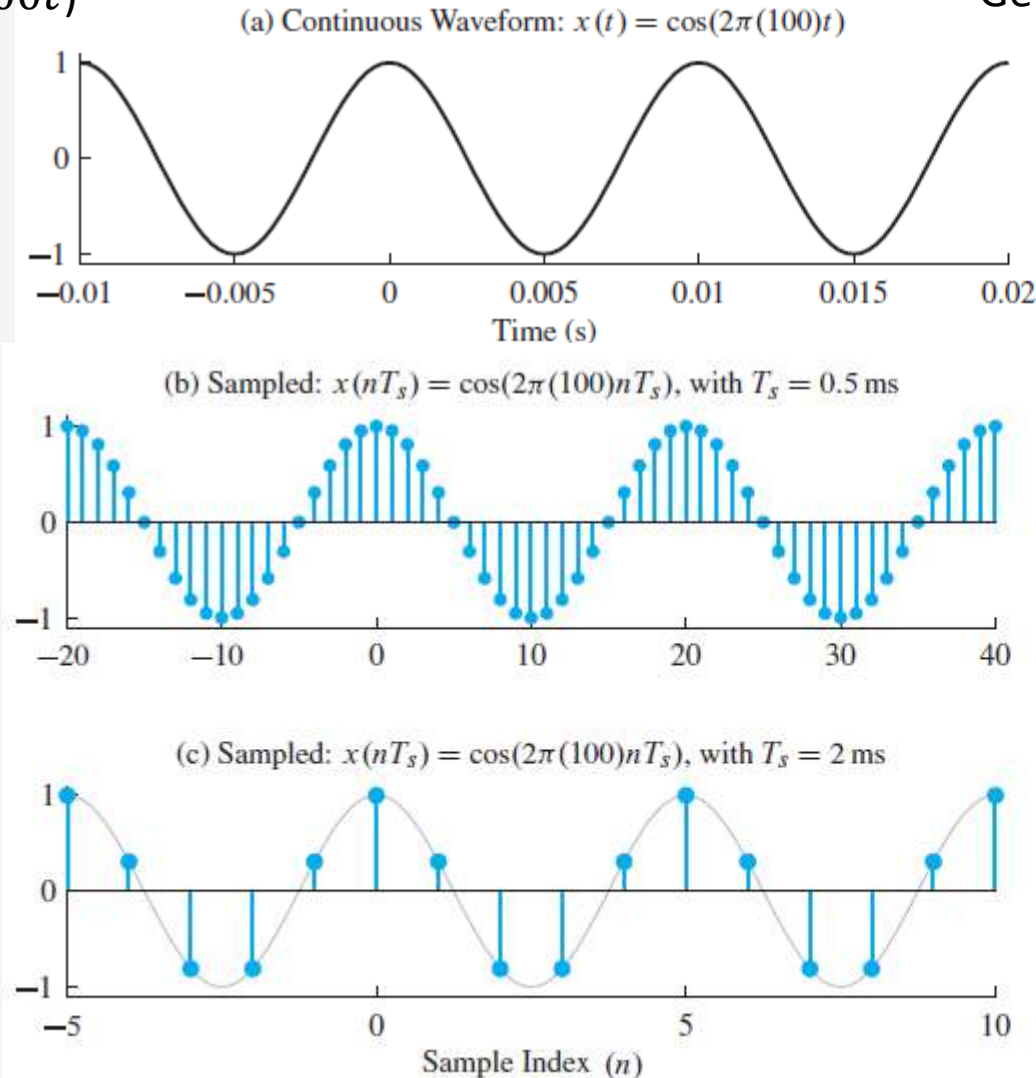
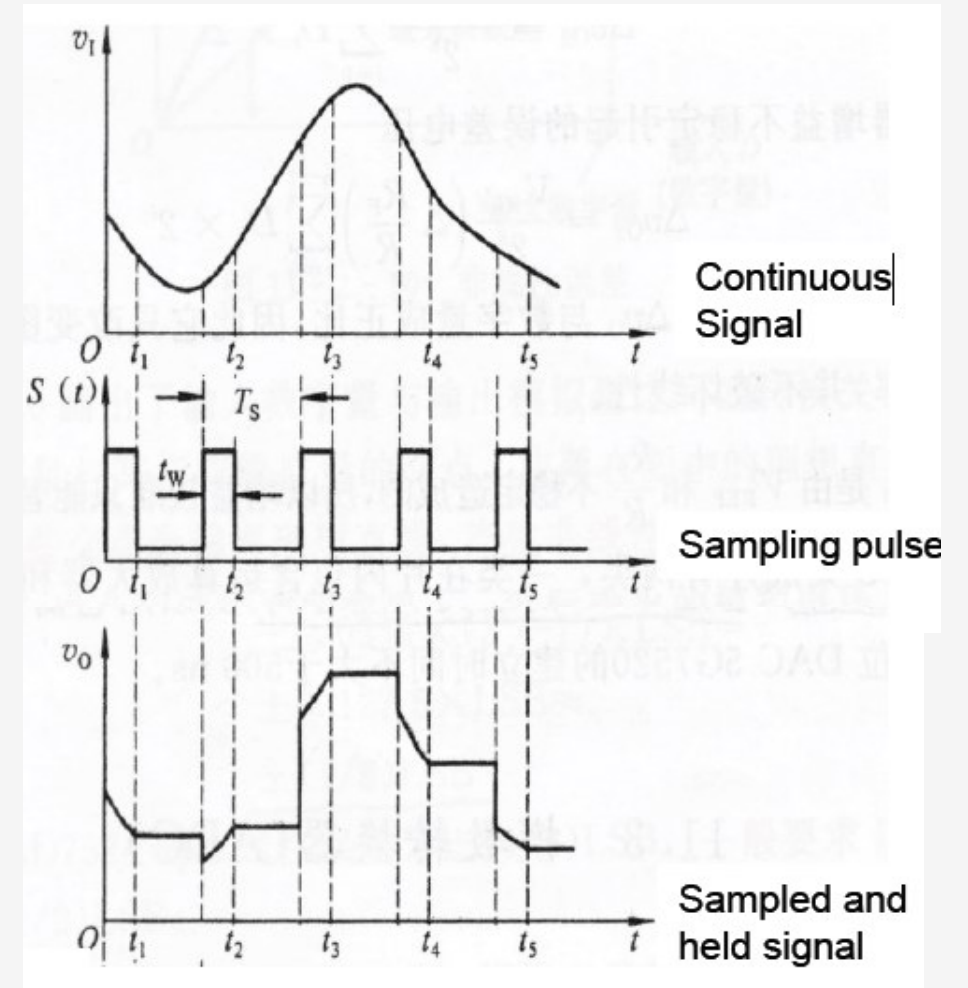
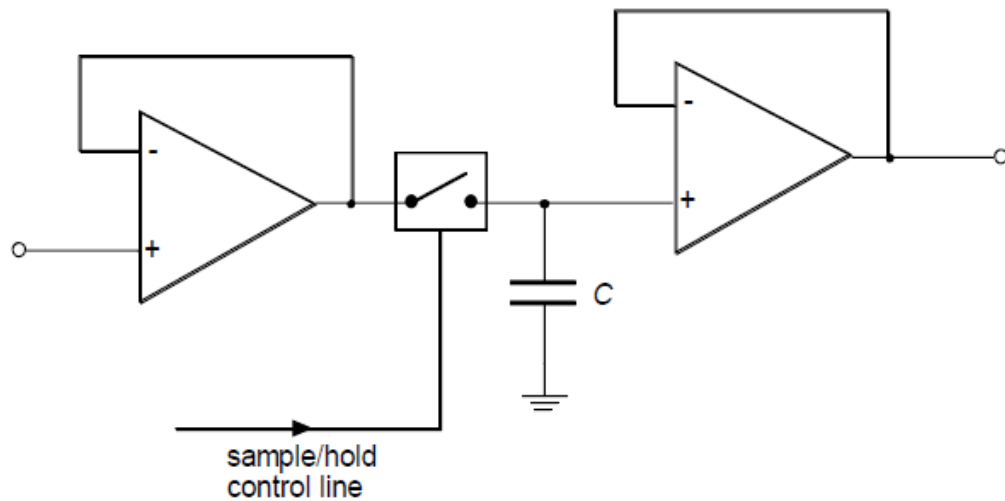


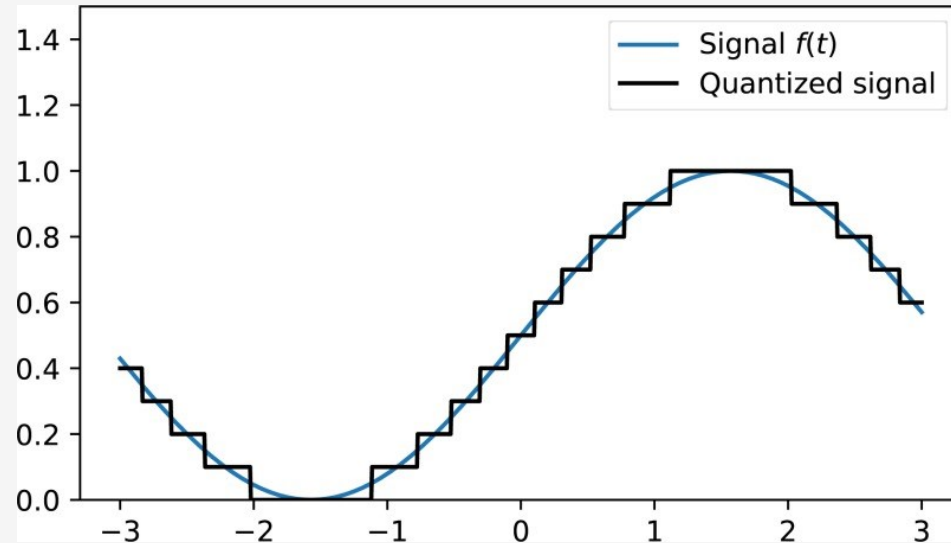
Figure 4-3 A continuous-time 100 Hz sinusoid (a) and two discrete-time sinusoids formed by sampling at $f_s = 2000$ samples/s (b) and at $f_s = 500$ samples/s (c).

Pratikte Örnekleme

- Load placed on the input of the circuit by charging the capacitor during the sample phase.
- Current flowing from the capacitor used in the conversion will reduce the voltage stored on the capacitor



Nicemleme (Quantization)



Çözünürlük:

Sayısal sinyale dönüştürürken sonuca etki eden en küçük analog sinyalin değişimi

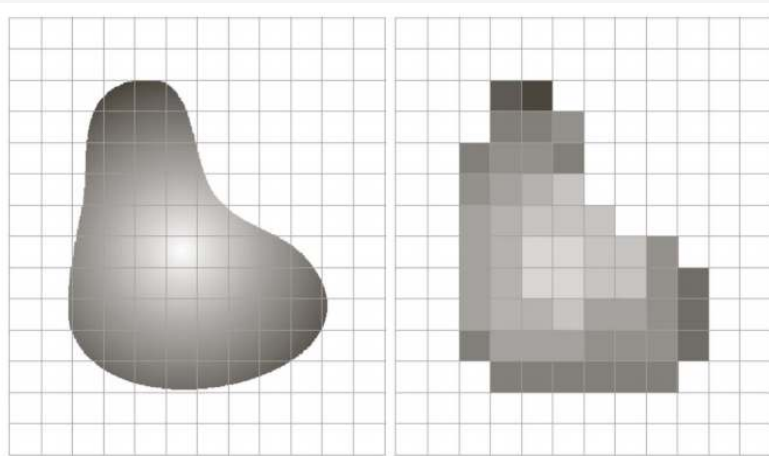
$$\Delta V = \frac{V_r}{2^N}$$

V = Reference voltage range

N = Number of bits in digital output.

2^N = Number of states.

ΔV = Resolution



a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

Çözünürlük terimi nicemleme hatasıyla doğrudan ilişkilidir.

Problems about Resolution



Example 1: Temperature range of 0 K to 300 K to be linearly converted to a voltage signal of 0 to 2.5 V, then digitized with an 8-bit A/D converter.

$2.5 / 2^8 = 0.0098$ V, or about 10 mV per step

$300 \text{ K} / 2^8 = 1.2$ K per step

Example-2:

- Assumes the input analog voltage is changing between 0-5 V.
- Using a 3-bit A/D converter draw the output as the input signal ramps from 0 to 5V.
- Calculate the resolution in volts. $5 / 2^3 \text{ V}$
- What is the maximum possible voltage out? (this is called the full-scale output) $(5 - \text{Resolution})$
- If the output is 011, what is the input? $3 \times 5 / 8 \text{ Volt}$

Nicemlemenin Sese Etkisi

8 bit seviyesi:



3 bit seviyesi:

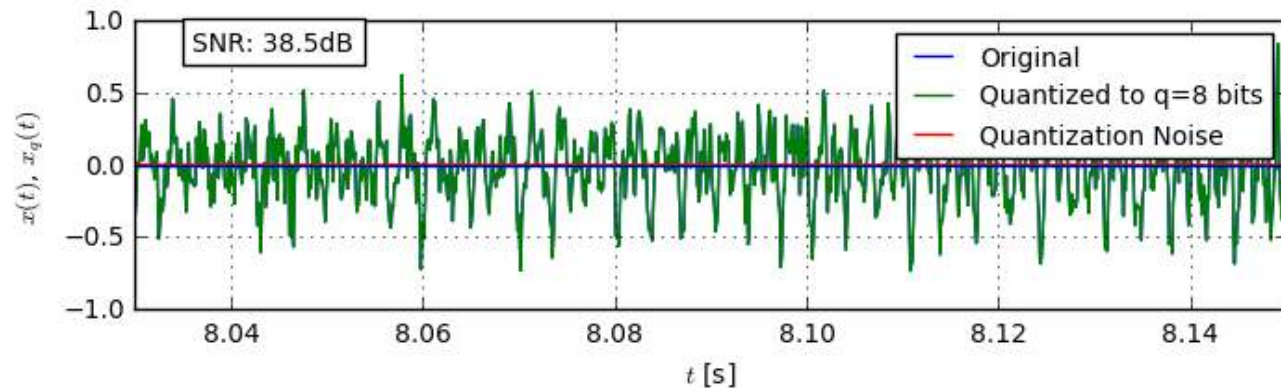
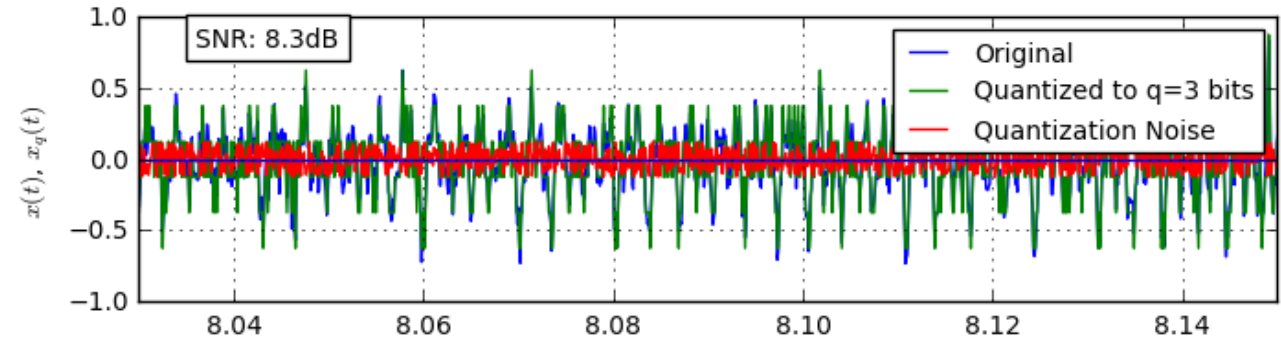
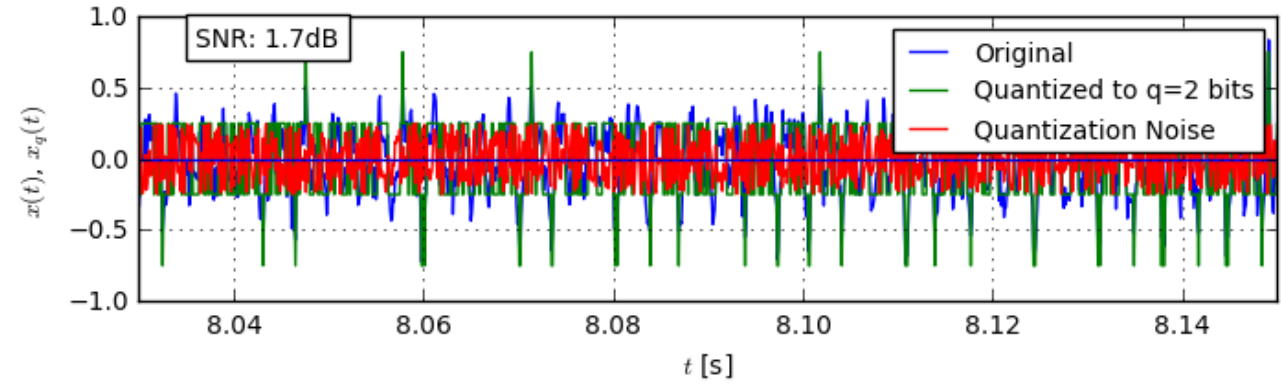


2 bit seviyesi:



Alternatif çözüm:

Non-uniform quantizier

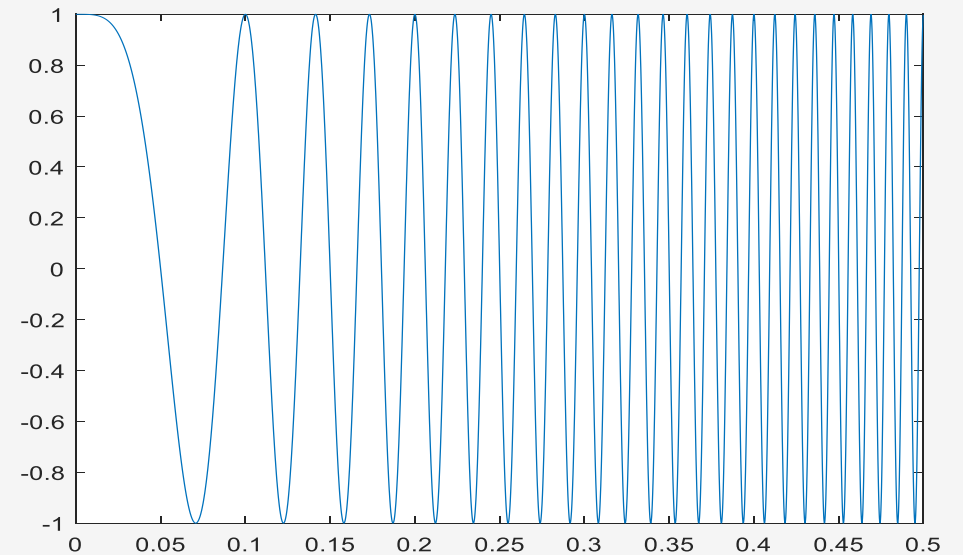


Let take a look at quantized version of Chirp Signal



A chirp is a signal in which the frequency increases (up-chirp) or decreases (down-chirp) with time

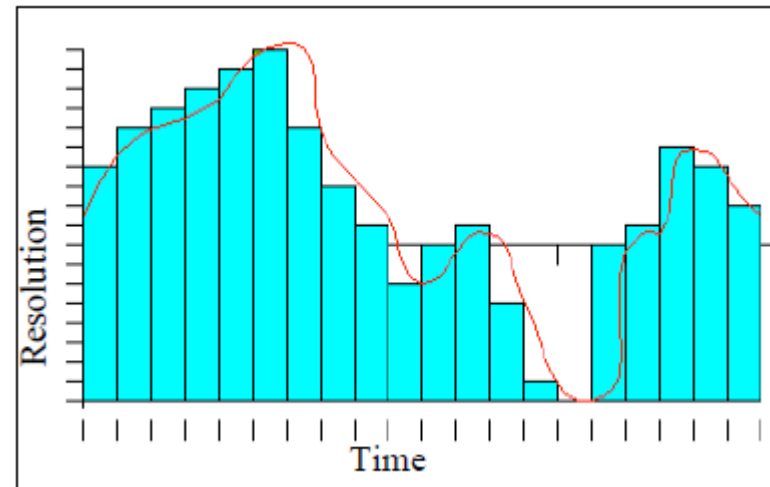
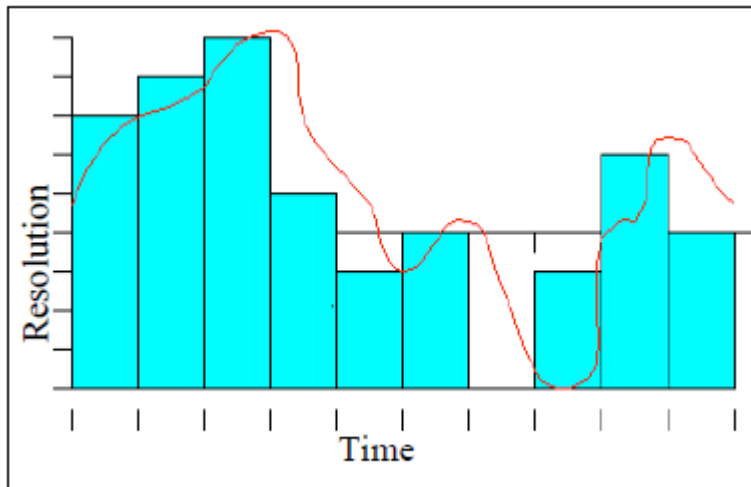
```
t = 0:1/8000:5;  
y = cos(2*pi*100*(t.*t+1));  
  
Qlevel = 2;  
Qstep = (1.01 - (-1))/Qlevel;  
  
Q = floor(y/Qstep)*Qstep + Qstep/2;  
  
figure(1), plot(t(1:4000),y(1:4000));  
figure(2), plot(t(1:4000),Q(1:4000));  
  
sound(y);  
sound(Q);
```



Accuracy of ADCs

There are two ways to best improve the accuracy of A/D conversion:

- increasing the resolution which improves the accuracy in measuring the amplitude of the analog signal.
- increasing the sampling rate which increases the maximum frequency that can be measured.



ADC Comparison

Product Attribute	Attribute Value
Manufacturer:	Analog Devices Inc.
Product Category:	Analog to Digital Converters - ADC
RoHS:	 Details
Series:	AD7641
Mounting Style:	SMD/SMT
Package / Case:	LFCSP-48
Resolution:	18 bit
Number of Channels:	1 Channel
Sampling Rate:	2 MS/s
Input Type:	Differential
Interface Type:	Parallel, Serial, 2-Wire, DSP, SPI
Architecture:	SAR
Reference Type:	External, Internal
SNR - Signal to Noise Ratio:	93.5 dB
Minimum Operating Temperature:	- 40 C
Maximum Operating Temperature:	+ 85 C

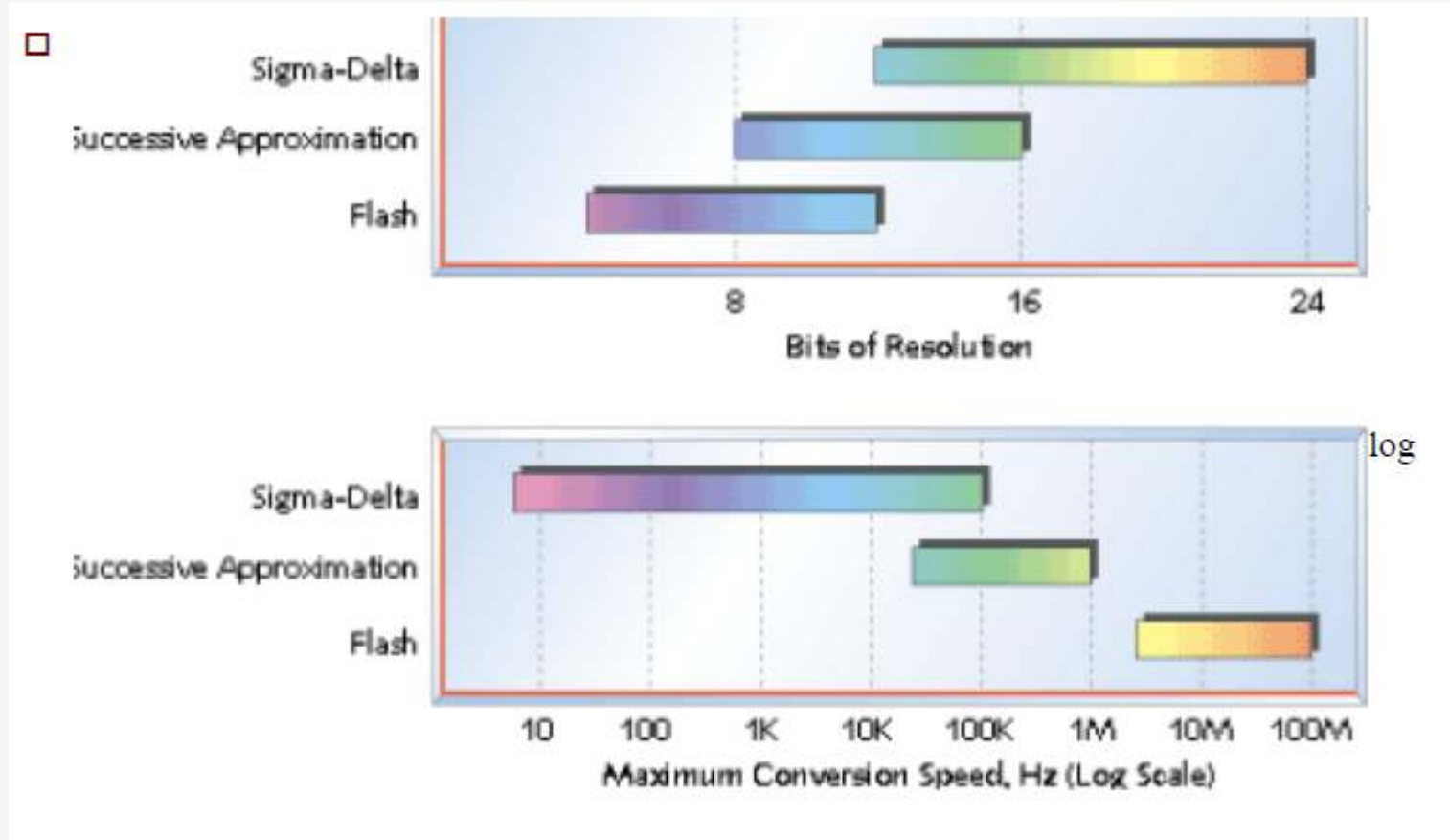


Product Attribute	Attribute Value
Manufacturer:	Texas Instruments
Product Category:	Analog to Digital Converters - ADC
RoHS:	 Details
Series:	TLV0832
Mounting Style:	Through Hole
Package / Case:	PDIP-8
Resolution:	8 bit
Number of Channels:	2 Channel
Sampling Rate:	44.7 kS/s
Input Type:	Differential/Single-Ended
Interface Type:	Serial
Architecture:	SAR
Reference Type:	Supply
Analog Supply Voltage:	2.7 V to 3.6 V
Digital Supply Voltage:	2.7 V to 3.6 V
Minimum Operating Temperature:	- 40 C
Maximum Operating Temperature:	+ 85 C

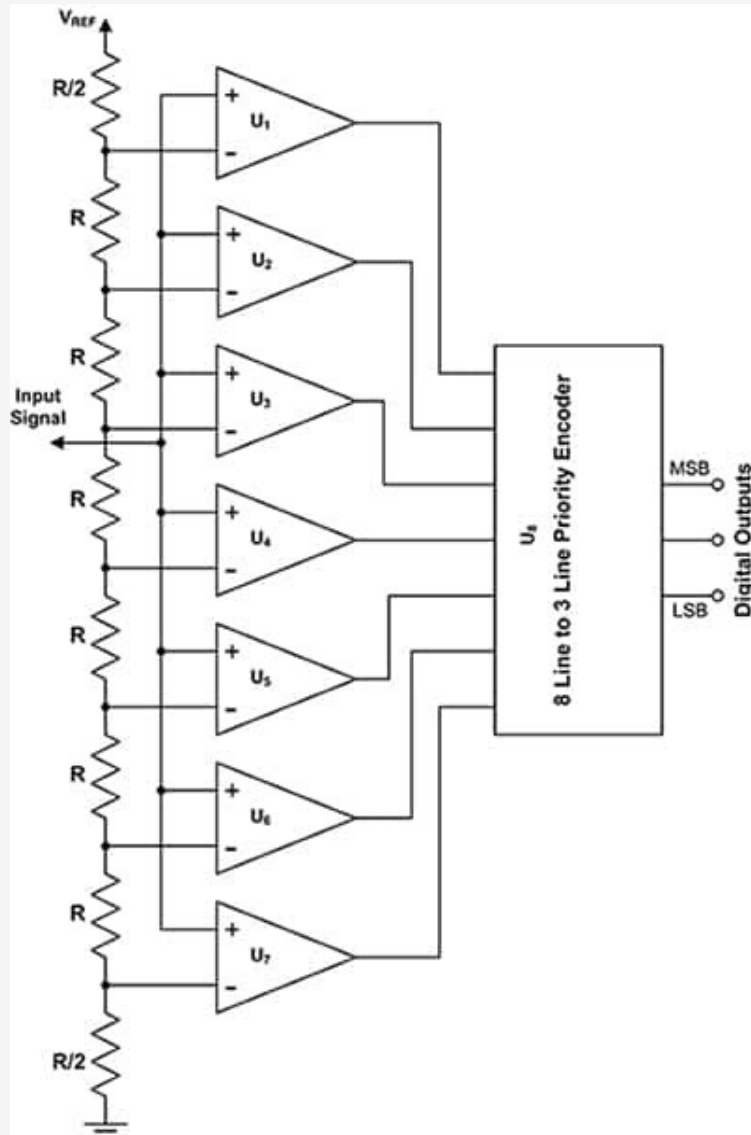


Types of A/D Converters

- Delta-Sigma ADCs
- Flash ADCs
- Successive Approximation ADCs
- Dual Slope ADCs
- Others...

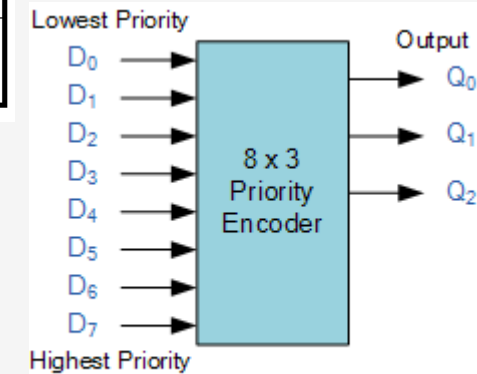


Flash ADC



If	Output
$V_{IN} > V_{REF}$	High
$V_{IN} < V_{REF}$	Low

- Requires 4095 parallel comparators for 12-bit.



Inputs								Outputs		
D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀	Q ₂	Q ₁	Q ₀
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	x	0	0	1
0	0	0	0	0	1	x	x	0	1	0
0	0	0	0	1	x	x	x	0	1	1
0	0	0	1	x	x	x	x	1	0	0
0	0	1	x	x	x	x	x	1	0	1
0	1	x	x	x	x	x	x	1	1	0
1	x	x	x	x	x	x	x	1	1	1

X = don't care

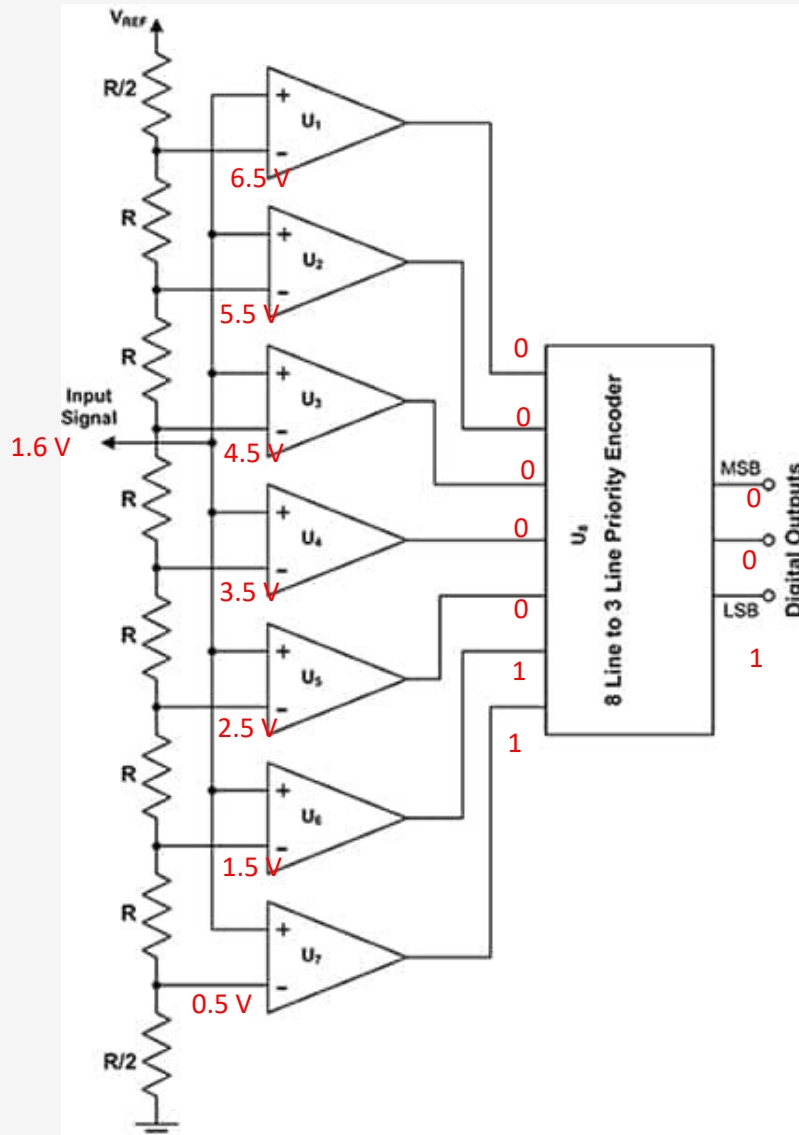
PROS

- Very Fast (Fastest)
- Very simple operational theory
- Speed is only limited by gate and comparator propagation delay

CONS

- Expensive
- Prone to produce glitches in the output
- Each additional bit of resolution requires twice the comparators.

Example: Flash ADC



Problem: If $V_{ref} = 8\text{ Volt}$ and $V_{in} = 1.6\text{ Volt}$ then find digital outputs.

Encoder Input : 1 1 0 0 0 0 0

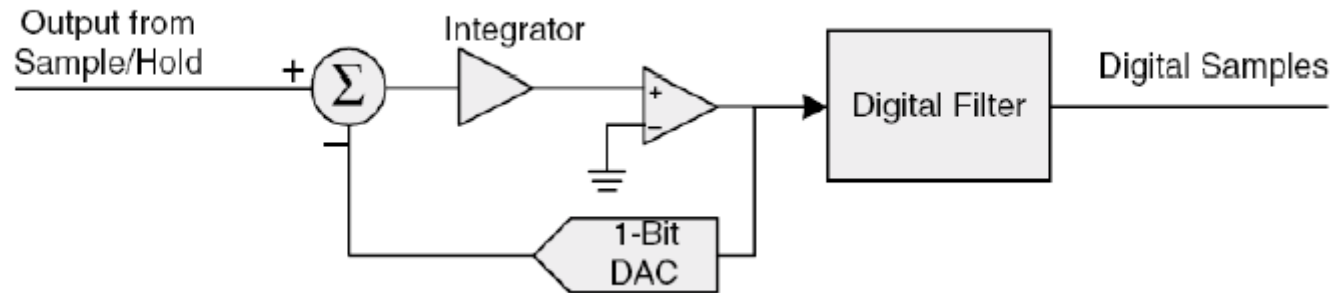
Digital Outputs, LSB to MSB : 0 0 1

Problem-2: If $V_{ref} = 8\text{ Volt}$ and $V_{in} = 8\text{ Volt}$ then find digital outputs.

Encoder Input : 1 1 1 1 1 1 1

Digital Outputs, LSB to MSB : 1 1 1

Sigma-Delta ADC



Input is over sampled, and goes to integrator.

The integration is then compared to ground.

Iterates and produces a serial bit stream

Output is a serial bit stream with # of 1's

PROS

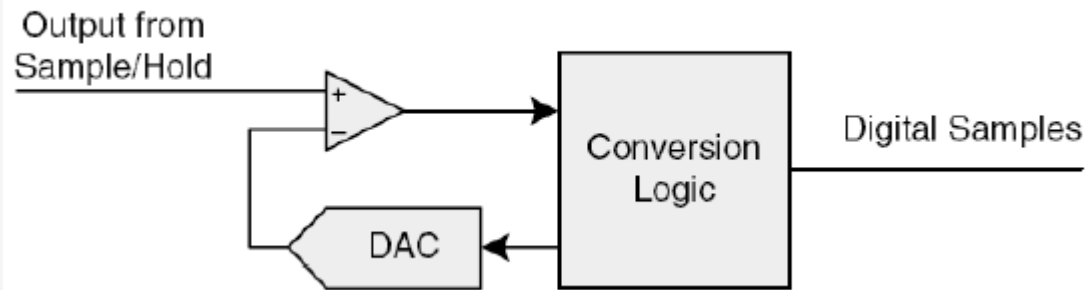
- High Resolution
- No need for precision components

CONS

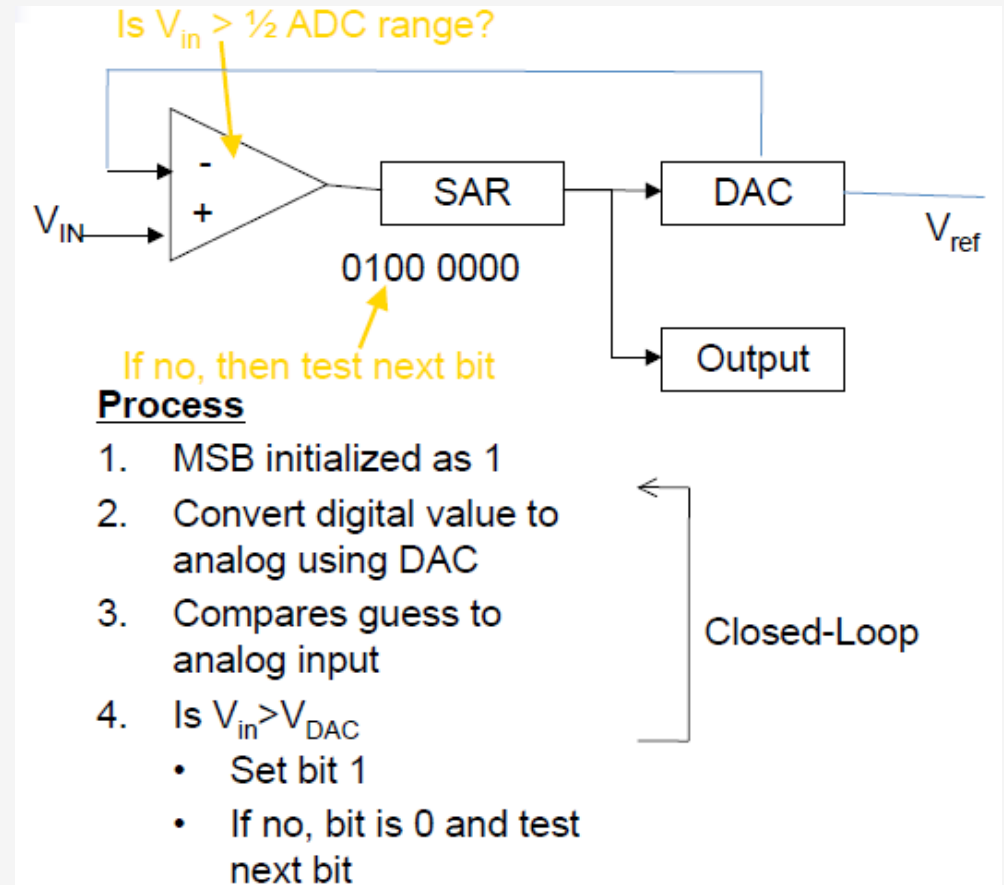
- Slow due to over sampling
- Only good for low bandwidth

<https://www.analog.com/en/design-center/interactive-design-tools/sigma-delta-adc-tutorial.html>

Successive Approximation (SAR)

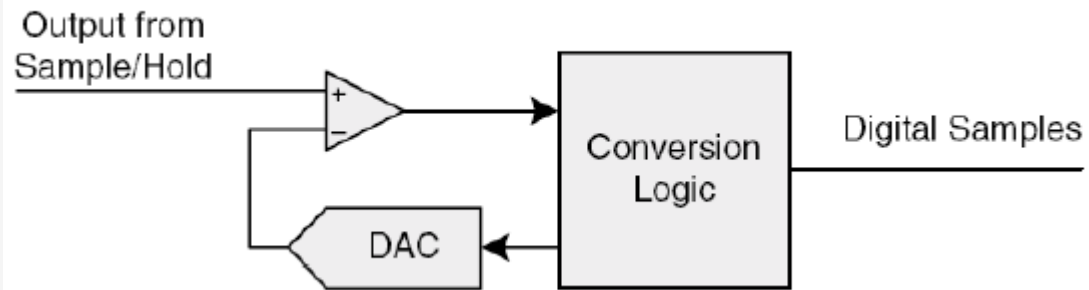


- Uses an internal n-bit DAC
- Conversion logic is a simple n-bit counter
- N-bit ADC requires 2^n cycles to perform a conversion in worst case.

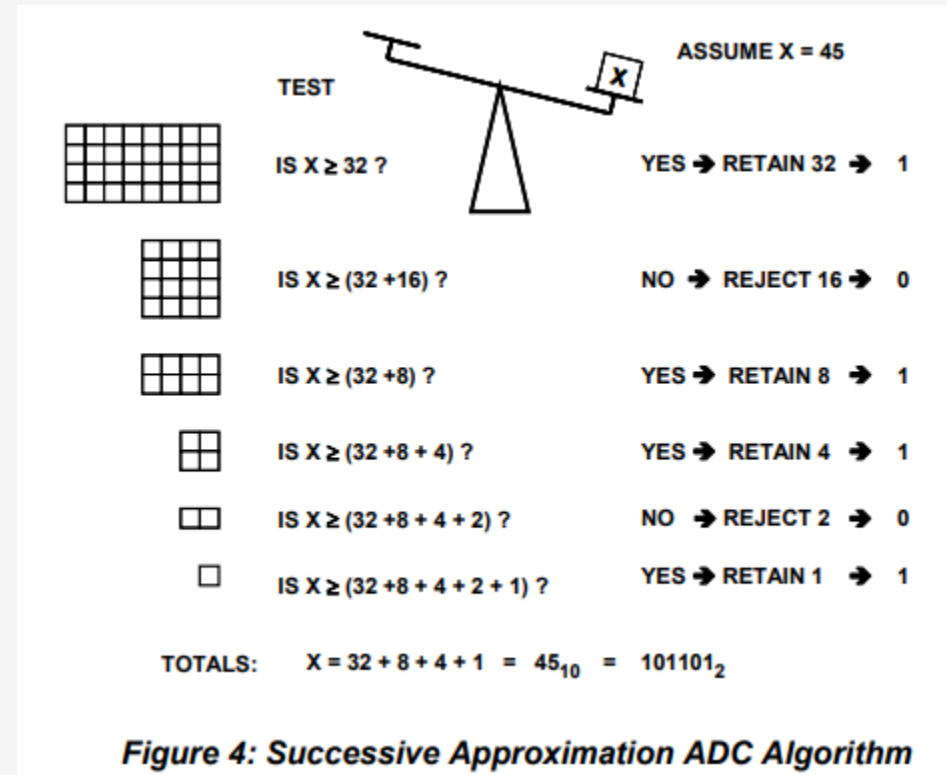


Uses a closed-loop to obtain best approximation.

Successive Approximation (SAR)

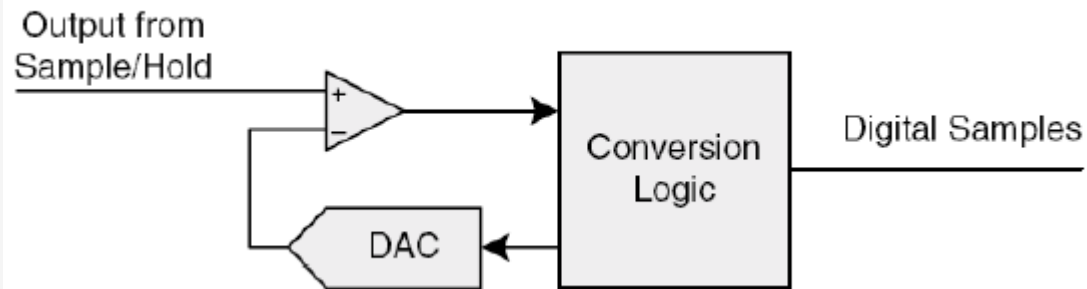


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Successive Approximation (SAR)



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- N-bit ADC requires 2^n cycles to perform a conversion in worst case.

Advantages

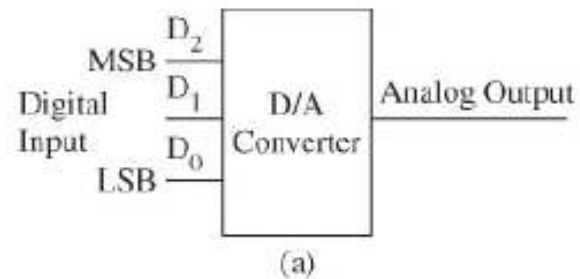
- Capable of high speed and reliable
- Medium accuracy compared to other ADC types
- Good tradeoff between speed and cost
- Capable of outputting the binary number in serial (one bit at a time) format.

Disadvantages

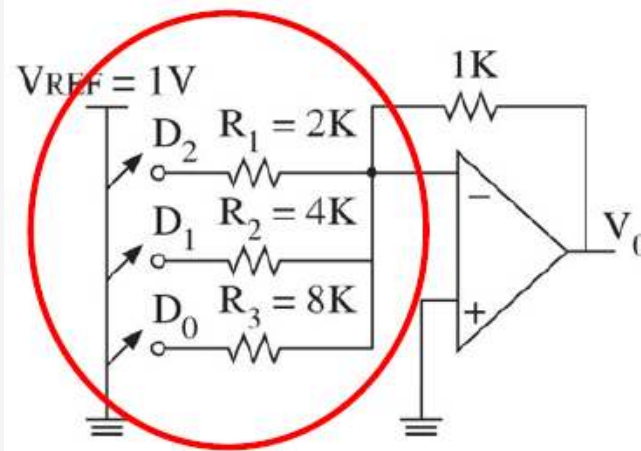
- Higher resolution successive approximation ADC's will be slower
- Speed limited to ~5Mps

Uses a closed-loop to obtain best approximation.

Digital to Analog Conversion

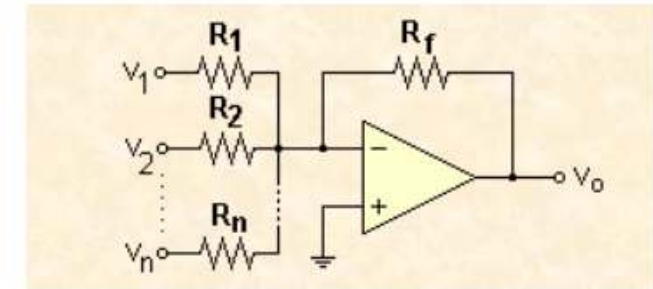


3-bit D/A Converter Circuit



Summing amplifier

R/2R Ladder Network for D/A Converter



The transfer function of the summing amplifier :
$$v_o = -(v_1/R_1 + v_2/R_2 + \dots + v_n/R_n)R_f$$

Thus if all input resistors are equal, the output is a scaled sum of all inputs.
If they are different, the output is a **weighted** linear sum of all inputs.

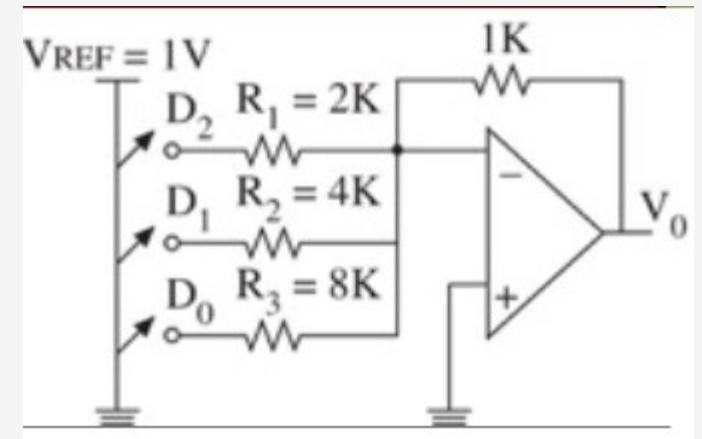
Example Case

- If the reference voltage is 1 V, and if all switches are connected, the output current can be calculated as follows:

$$I_o = I_T = I_1 + I_2 + I_3 = \frac{V_{REF}}{R_1} + \frac{V_{REF}}{R_2} + \frac{V_{REF}}{R_3} = \frac{V_{REF}}{1k} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) = 0.875 \text{ mA}$$

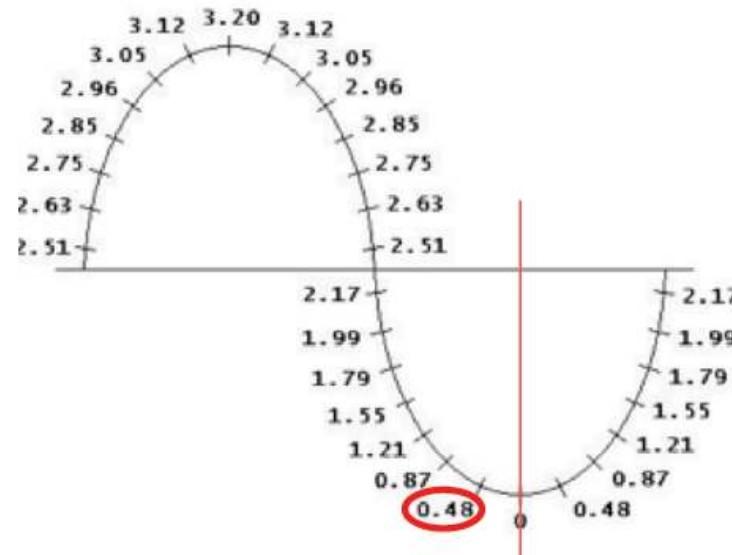
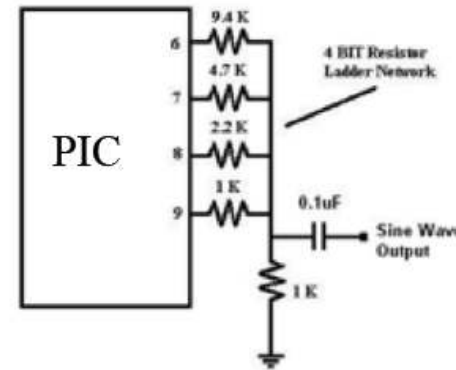
- Output voltage

$$V_o = -R_f I_T = -(1k) \times (0.875 \text{ mA}) = -0.875 \text{ V} = \left| \frac{7}{8} \text{ V} \right|$$



Can we generate a sinusoidal signal?

- Theoretically the voltages would range from 0 to 5
- How do you change the frequency?



Voltage Out for BIT combinations		
Sine D Angle	Pins 9 8 7 6	Vout
270	0000	0
258.75, 281.25	1000	0.48
247.50, 292.50	0100	0.87
236.25, 303.75	1100	1.21
225.00, 315.00	0010	1.55
213.75, 326.25	1010	1.79
202.50, 337.50	0110	1.99
191.25, 348.75	1110	2.17
11.25, 168.75	0001	2.51
22.50, 157.50	1001	2.63
33.75, 146.25	0101	2.75
45.00, 135.00	1101	2.85
56.25, 123.75	0011	2.96
67.50, 112.50	1011	3.05
78.75, 101.25	0111	3.12
90	1111	3.20

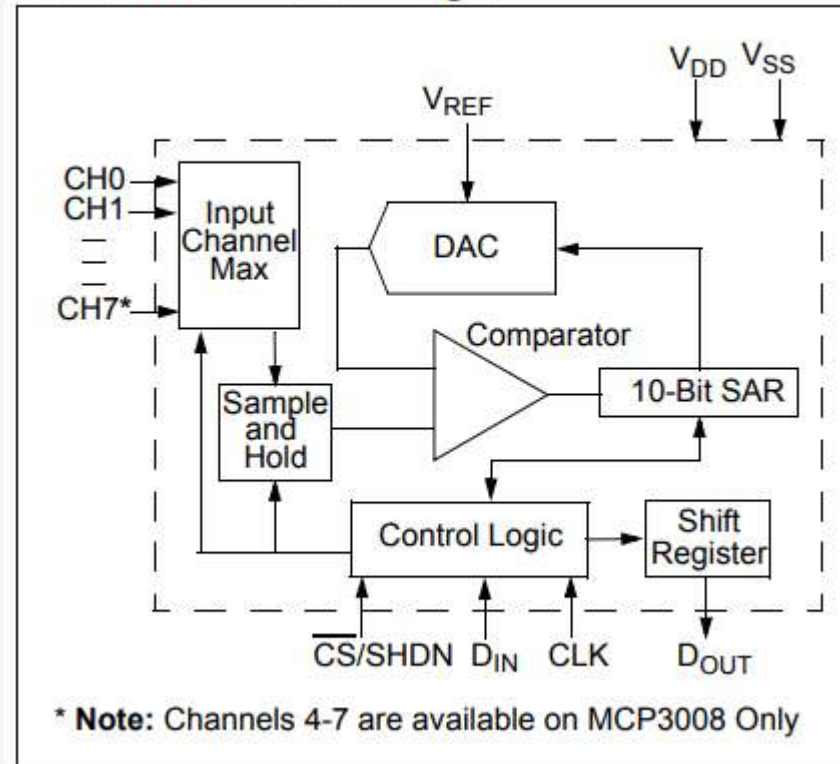
Example: MCP3008 ADC - Adafruit



Features

- 10-bit resolution
- ± 1 LSB max DNL
- ± 1 LSB max INL
- 4 (MCP3004) or 8 (MCP3008) input channels
- Analog inputs programmable as single-ended or pseudo-differential pairs
- On-chip sample and hold
- SPI serial interface (modes 0,0 and 1,1)
- Single supply operation: 2.7V - 5.5V
- 200 ksps max. sampling rate at $V_{DD} = 5V$
- 75 ksps max. sampling rate at $V_{DD} = 2.7V$
- Low power CMOS technology
- 5 nA typical standby current, 2 μA max.
- 500 μA max. active current at 5V
- Industrial temp range: $-40^{\circ}C$ to $+85^{\circ}C$
- Available in PDIP, SOIC and TSSOP packages

Functional Block Diagram



CH0	1	16	V _{DD}
CH1	2	15	V _{REF}
CH2	3	14	AGND
CH3	4	13	CLK
CH4	5	12	D _{OUT}
CH5	6	11	D _{IN}
CH6	7	10	CS/SHDN
CH7	8	9	DGND

DIN: Serial Data In
DOUT: Serial Data Out
CLK: Serial Clock
CS: Chip Select

CHX: Analog Input for input X

Example – 1: Measuring a Voltage

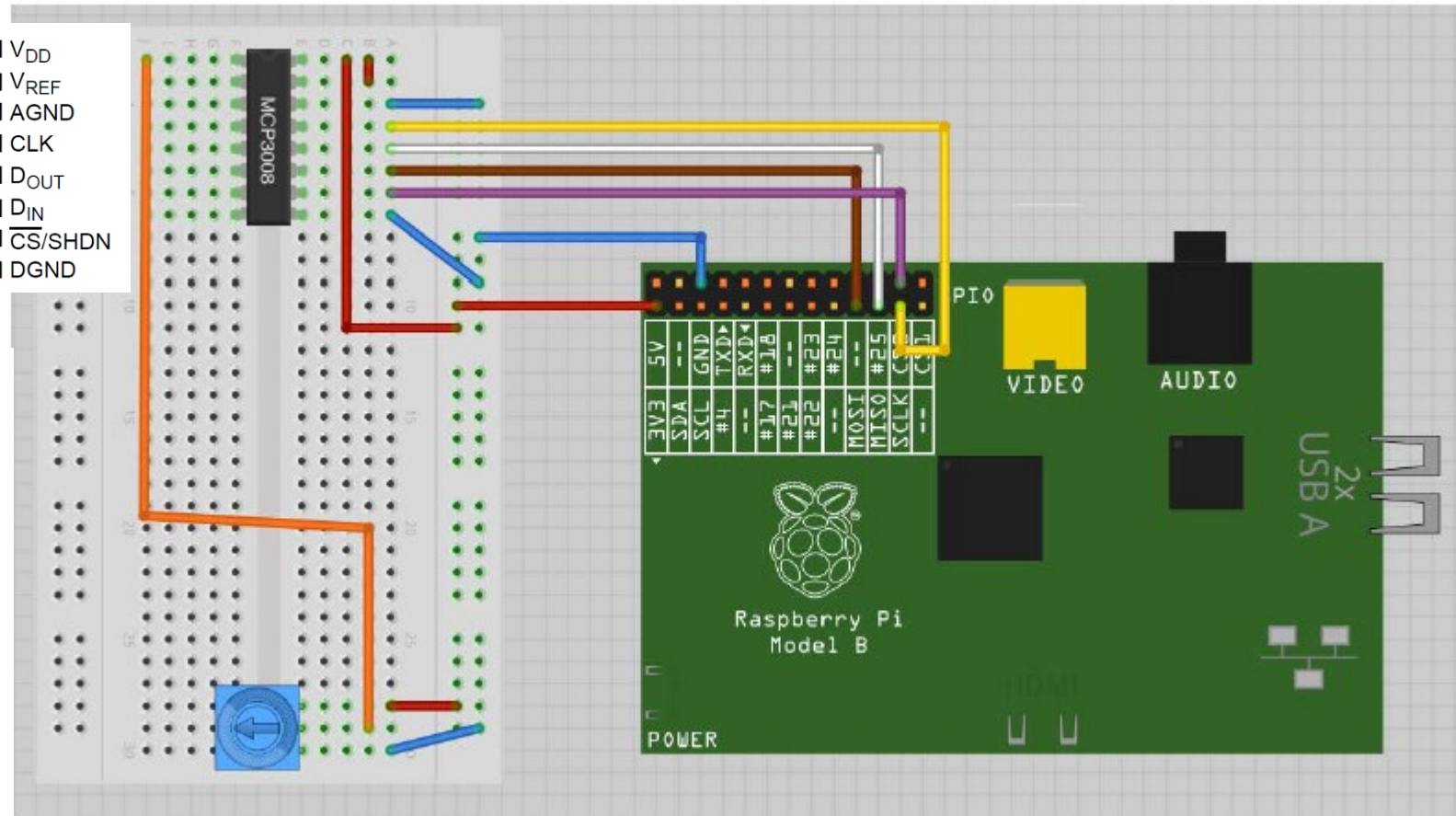
CH0	1	16	V _{DD}
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CH6	7	10	CS/SHDN
CH7	8	9	DGND

```
import spidev, time
```

```
spi = spidev.SpiDev()  
spi.open(0, 0)
```

```
def analog_read(channel):  
    r = spi.xfer2([1, (8 + channel) << 4, 0])  
    adc_out = ((r[1]&3) << 8) + r[2]  
    return adc_out
```

```
while True:  
    reading = analog_read(0)  
    voltage = reading * 3.3 / 1024  
    print("Reading=%d\tVoltage=%f" % (reading, voltage))  
    time.sleep(1)
```



Example – 1: Measuring a Voltage



```
import spidev, time

spi = spidev.SpiDev()
spi.open(0, 0)

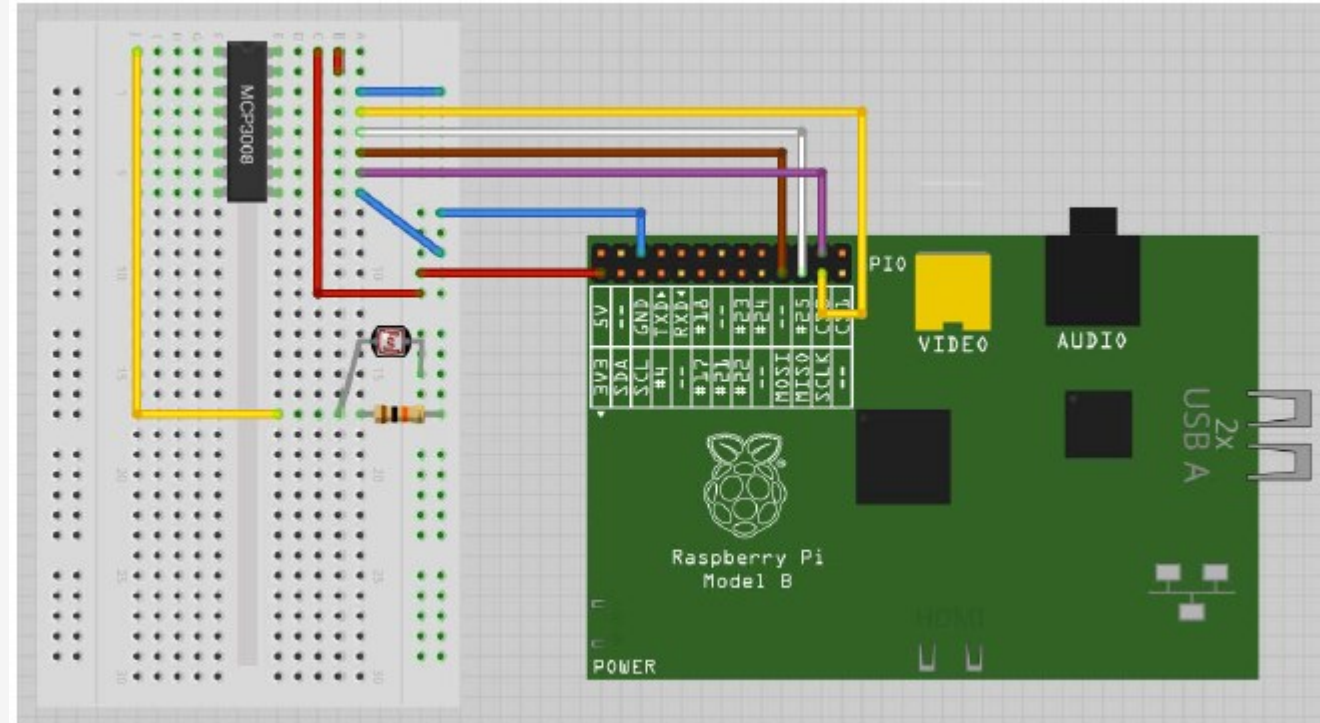
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while True:
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    voltage = reading * 3.3 / 1024
    print("Reading=%d\tVoltage=%f" % (reading, voltage))
    time.sleep(1)
```

```
$ sudo python adc_test.py
Reading=0          Voltage=0.000000
Reading=126        Voltage=0.406055
Reading=221        Voltage=0.712207
Reading=305        Voltage=0.982910
Reading=431        Voltage=1.388965
Reading=527        Voltage=1.698340
Reading=724        Voltage=2.333203
Reading=927        Voltage=2.987402
Reading=10         Voltage=3.296777
Reading=1020       Voltage=3.287109
Reading=1022       Voltage=3.293555
```

Example 2 – Measuring the Darkness

LDR: Photoresistor



Example 3 – Using Triple-Axis Accelerometer

ADXL335 triple-axis accelerometer

It uses three channels of the ADC to measure the X, Y, and Z acceleration forces.

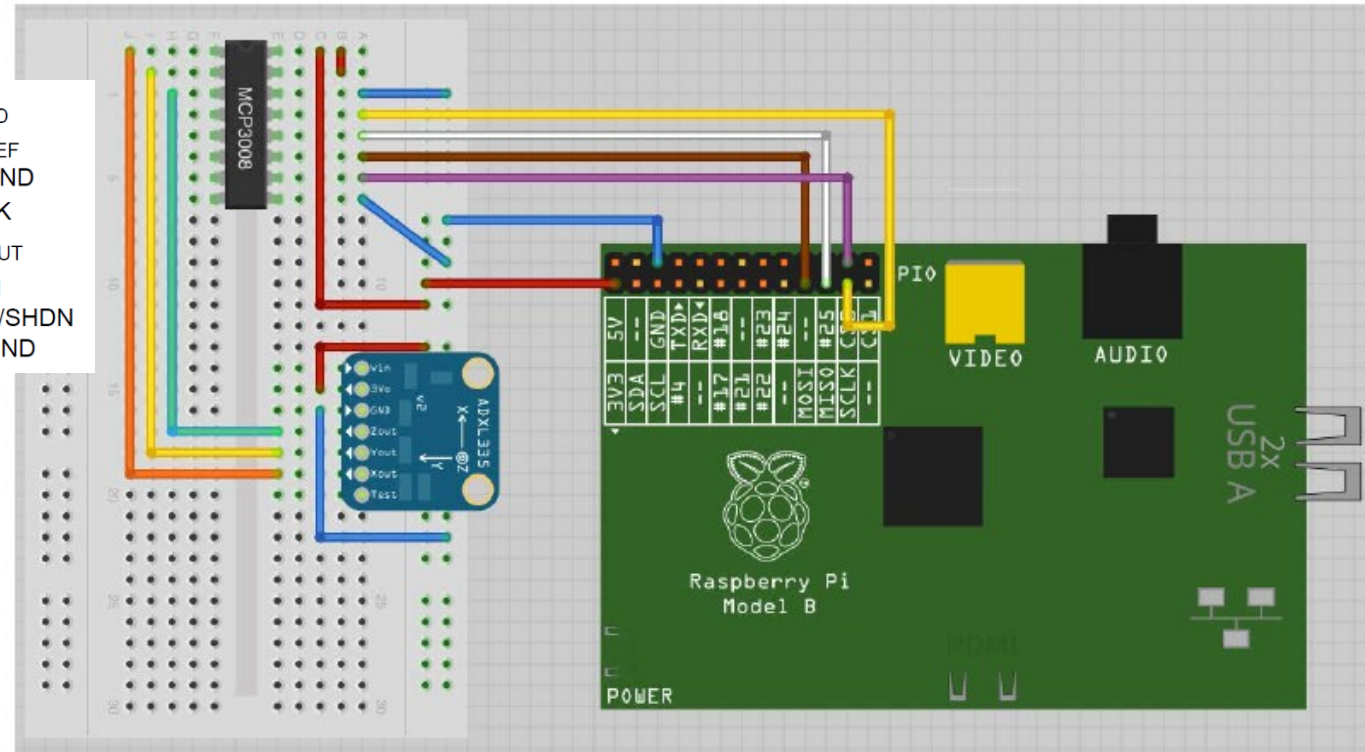
CH0	1	16	V _{DD}
CH1	2	15	V _{REF}
CH2	3	14	AGND
CH3	4	13	CLK
CH4	5	12	D _{OUT}
CH5	6	11	D _{IN}
CH6	7	10	CS/SHDN
CH7	8	9	DGND

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spi.open(0,0)

def analog_read(channel):
    r = spi.xfer2([1, (8 + channel) << 4, 0])
    adc_out = ((r[1]&3) << 8) + r[2]
    return adc_out

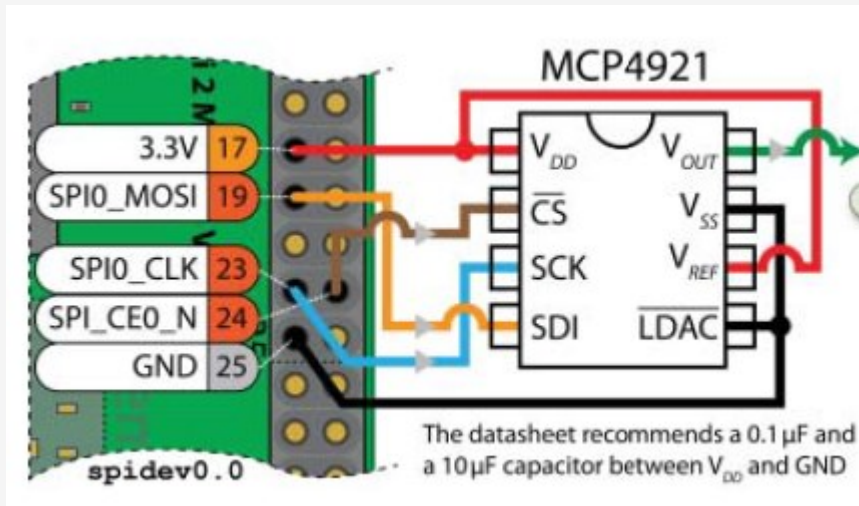
while True:
    x = analog_read(0)
    y = analog_read(1)
    z = analog_read(2)
    print("X=%d\tY=%d\tZ=%d" % (x, y, z))
    time.sleep(1)
```



Example – 4: SPI DAC with C program

The MCP4921 is a low-cost (\$2) single-channel 12-bit SPI DAC

The DAC does not send data back to the RPi, so there is no MISO connection required.



```
#include <iostream>
#include <math.h>
#include "bus/SPIDevice.h"
using namespace exploringRPi;

int main() { // mask = (MSB) 0 (BUF) 0 (GA) 1 (SHDN) 1
    // Red box
    // Blue box
    busDevice->close();
    std::cout << "End of RPi SPI DAC Example" << std::endl;
    return 0;
}
```